

Assignment 2 Report

Group 17:
Darlington Nkrumah (202492437)
Greg de Souza (202582225)
Xuan Toan Doan (202583882)

February 2026

Introduction

For assignment 2 we were provided a two data sets, labeled “train.csv” and “test.csv” containing 8 features measuring some property of concrete, and the value of the concrete compressive strength. For the sake of readability, the 8 features will be labeled as x_i according to the table 1, and the target variable (Compressive Strength) will be y . Vectors will be written with bold uppercase letters $\mathbf{X}$ for a feature vector, and θ for a set of parameters, with θ_0 reserved for the intercept. The collection of feature vectors from a data set will be labeled $\mathcal{D}$.

In all questions the quality of fit was accessed with the R^2 metric, and the Err with the RSE metric

Feature Name	Variable
<i>Cement_component1_kgInAM_3Mixture_</i>	x_1
<i>BlastFurnaceSlag_component2_kgInAM_3Mixture_</i>	x_2
<i>FlyAsh_component3_kgInAM_3Mixture_</i>	x_3
<i>Water_component4_kgInAM_3Mixture_</i>	x_4
<i>Superplasticizer_component5_kgInAM_3Mixture_</i>	x_5
<i>CoarseAggregate_component6_kgInAM_3Mixture_</i>	x_6
<i>FineAggregate_component7_kgInAM_3Mixture_</i>	x_7
<i>Age_day_</i>	x_8
<i>ConcreteCompressiveStrength_MPa_Megapascals_</i>	y

Table 1: Feature names with their symbolic variables

Question 1: Simple Linear Regression

The first question tasks us to train a predictor for y using an ordinary least squares with a traditional linear regression model, such as in equation 1. The model was assessed for generalized error Err in two different ways. First the training data set was split in training/validation/test, with 20% of the data being randomly selected to be the test data set. Since there are no hyperparameters involved, the model was trained on the remaining “Train + Validation” dataset, consisting of 640 data points, and assessed for Generalized Error (Err) against the remaining 160 data points of the test set. The results are $R^2 = 0.6045$ and $RSE = 11.5$, no variance is available since it's a single test dataset. The parameters fitted by this models are presented in the table 3.

$$\hat{y} = \theta_0 + \mathbf{X}^T \theta \quad (1)$$

The same model was also assessed with a cross validation approach with 5 folds over the “train.csv” dataset, meaning each folds had 160 data points, similar to the test dataset from the previous approach. The choice of the number of folds seems adequate due to the fact that the number of remaining training data points (640) still being way larger than the number of features (8). As an objective measure to validate this choice of number of folds, we have also evaluated the variance of the estimated Err in the cross validation. The results were $R^2 = 0.6270 \pm 0.0002$ and $RSE = 11.6 \pm 0.3$, the low variance suggests that the 5 fold split results in representative values for the Err metrics. Also, that's the split that maximizes the R^2 , as shown in figure 1

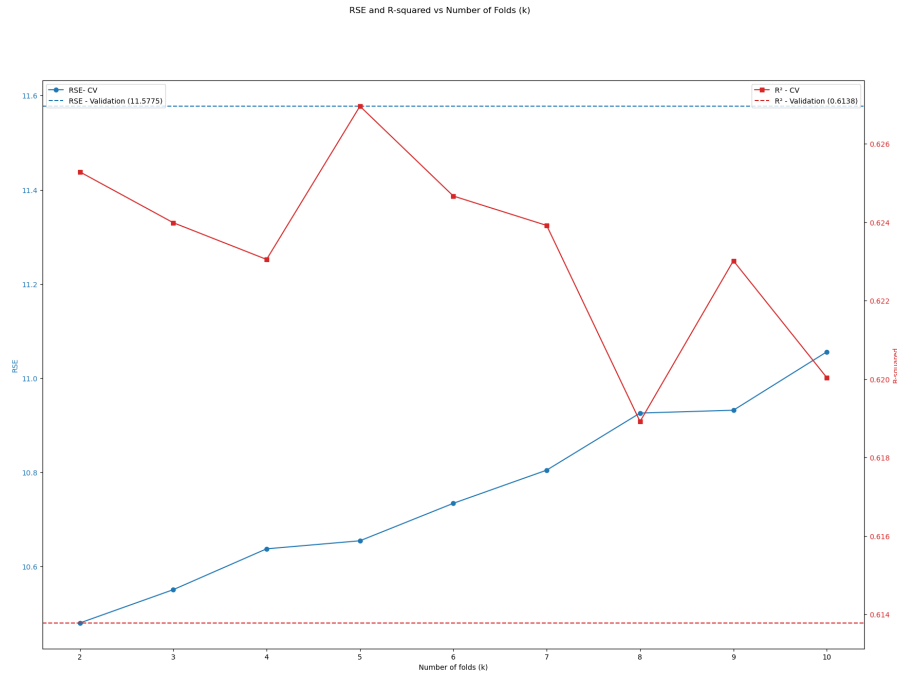


Figure 1: Assessment parameters given different number of folds for the Cross Validation Approach

	Split	CV
R^2	0.6045	0.6270 ± 0.0002
RSE	11.5	11.6 ± 0.3

Table 2: Results for the estimated quality of fit and Err using R^2 and RSE in the Train/Test split and in the Cross Validation Approach

The results for both the split and cross validation approaches is presented in table 2, while the results suggests neither method found an adequate fit, the cross validation approach accesses the model as worse than the split approach. This makes sense, as the with the cross validation we access the model multiple times with different folds, making this result are more trustworthy estimation of the generalization error, specially because we can also estimate the variance of the model.

In general, the bad performance is also expected in this “naive” approach, as the data features don’t “well behaved” in a linear sense, as can be seen in figure 4

Question 2: Ridge Regression

The second task involves fitting the same data using Ridge Regressions (equation 3) using the training dataset “train.csv”. For this task we employed Grid Search using cross validation with 5 folds, the search determined that the best value for the hyperparameter is $\alpha = 8111$, notably the data wasn’t regularized. Figure ?? presents the values of the RSE metric for different values of alpha, and there is a clear minimum around the found value.

$$\hat{y} = \theta_0 + \mathbf{X}^T \theta + \alpha \cdot \|\theta\|_2^2 \quad (2)$$

The **cross validation** assessment estimated the Err parameters as $R^2 = 0.6271$ and $RSE = 10.63$.

The model was then retrained using $\alpha = 8111$ on the total 800 data points of the training dataset, and accessed against the independent test dataset “test.csv”, resulting in $R^2 = 0.5952$ and $RSE = 10.00$. As expected the performance against unseen data was worse than evaluated by the the cross validation grid search.

For the same dataset and test, the ordinary least squares (OSL) result in a $R^2 = 0.5953$ and $RSE = 10.00$. So, given the variance estimated in question 1, both regression methods have as similar performance. This is consistent

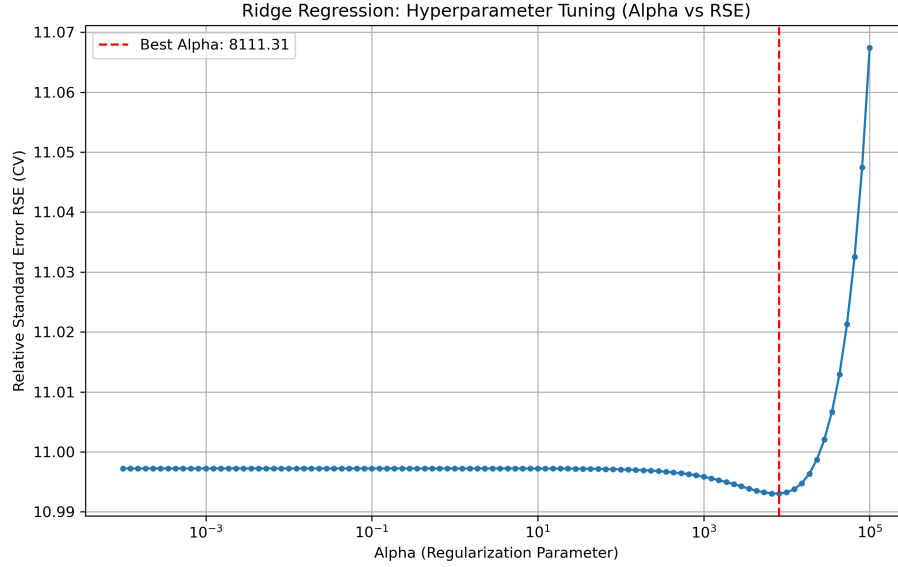


Figure 2: Assessment of RSE metric for different values of alpha for the Ridge Regression model

with figure 2, where a Ridge regression for α close to 0 seems to perform similar to the best performing α , it's reasonable to conclude that a “naive” linear relation between or targets and the features isn't enough to generate a good predictor.

Notably, while the performance is similar, the actual parameters found by Ridge Regression are significantly smaller than those found by ordinary regression, which makes sense, since the job of the parameter α is two limit the growth of the L2 metric of the parameter set. the best set of parameters obtained by Ridge Regressions is. The values of the final parameters are presented in table 3.

Parameter	Ordinary Least Squares	Ridge Regression	Lasso Regression
Intercept	-44.19	7.153	11.96
θ_1	0.1291	0.1169	0.1145
θ_2	0.1158	0.0994	0.0967
θ_3	0.1035	0.0858	0.0816
θ_4	-0.1328	-0.2163	-0.2213
θ_5	0.2056	0.1092	0.1553
θ_6	0.0245	0.0075	0.0065
θ_7	0.0296	0.0126	0.0099
θ_8	0.1253	0.1172	0.1178

Table 3: Final parameters fitted in Q1, Q2 and Q3 for Ordinary, Ridge and Lasso regression respectively

Question 3: Lasso Regression

The second task involves fitting the same data using Lasso Regressions (equation 3) using the training dataset “train.csv”. For this task we employed Grid Search using cross validation with 5 folds, the search determined that the best value for the hyperparameter is $\alpha = 0.0001$, notably the data wasn't regularized. As figure 3 shows, there is a plateau of equivalent vales of α between 0.0001 and 1, so any value of the hyperparameter would be perform similar to another. The **cross validation** assessment estimated the Err parameters as $R^2 = 0.6268$ and RSE= 10.63. The

$$\hat{y} = \theta_0 + \mathbf{X}^T \theta + \alpha \cdot \|\theta\|_2^2 \quad (3)$$

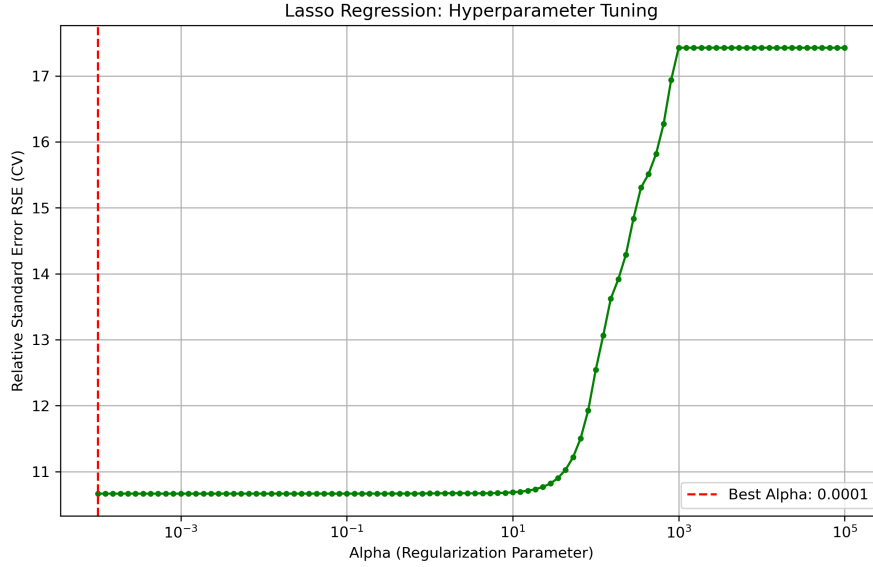


Figure 3: Assessment of RSE metric for different values of alpha for the Lasso Regression model

Err OLS	Ridge	Lasso	
R^2	0.5953	0.5952	0.5953
RSE	10.00	10.00	10.00

Table 4: Results for the estimated quality of fit and Err using R^2 and RSE in the Train/Test split and in the Cross Validation Approach

The model was then retrained using $\alpha = 0.0001$ on the total 800 data points of the training dataset, and accessed against the independent test dataset “test.csv”, resulting in $R^2 = 0.5953$ and $RSE = 10.00$. As expected the performance against unseen data was worse than evaluated by the the cross validation grid search.

Again, the performance is similar to the last two regression approaches, suggesting that the best performance for a linear relation between features and data is indeed around 0.6 for R^2 . But the Lasso regression also resulted in some values of the parameters being very close to 0, notably for θ_6 and θ_7 , suggesting that the respective features x_6 and x_7 don’t have meaningful linear relation with our target y .

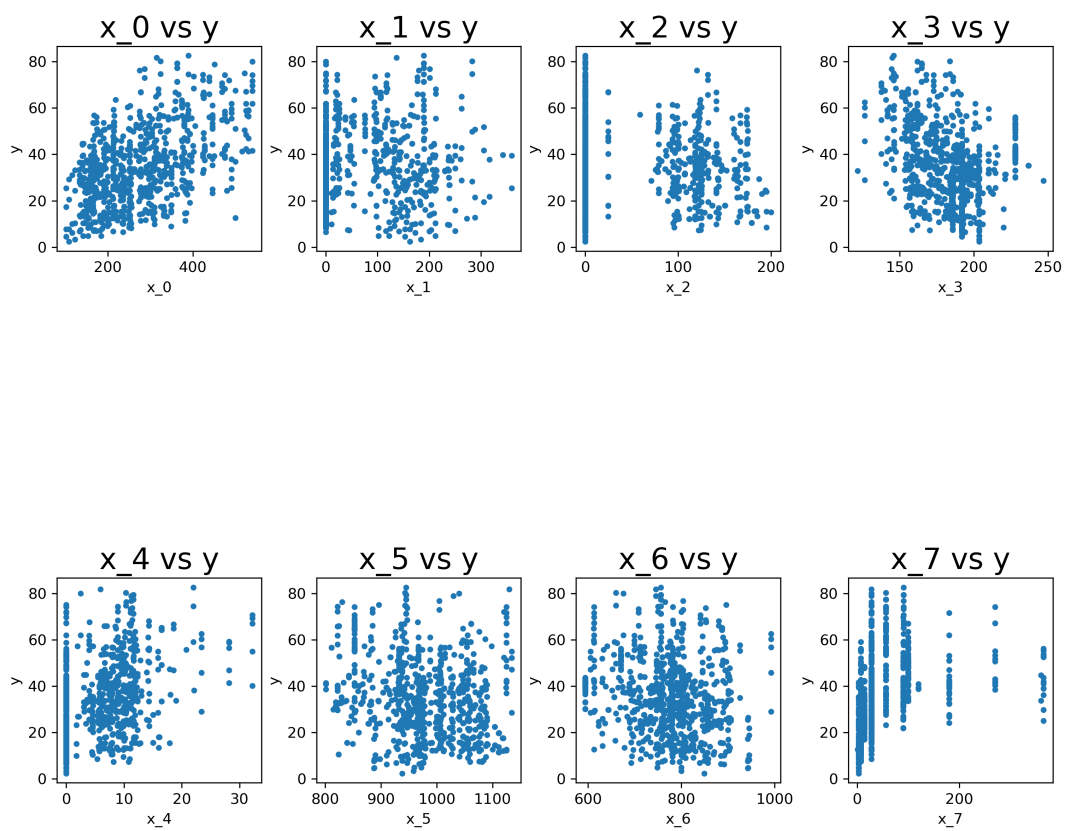


Figure 4: Features plotted against target